



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

PRESSURE INSTRUMENT PROCESS DATA SHEET

Docu. No. : 14780-8120-PS-0032
Project No. : 120117
Rev. No. : Z1
Date : MAY 10, 2016
Page : 1 / 40

TITLE :

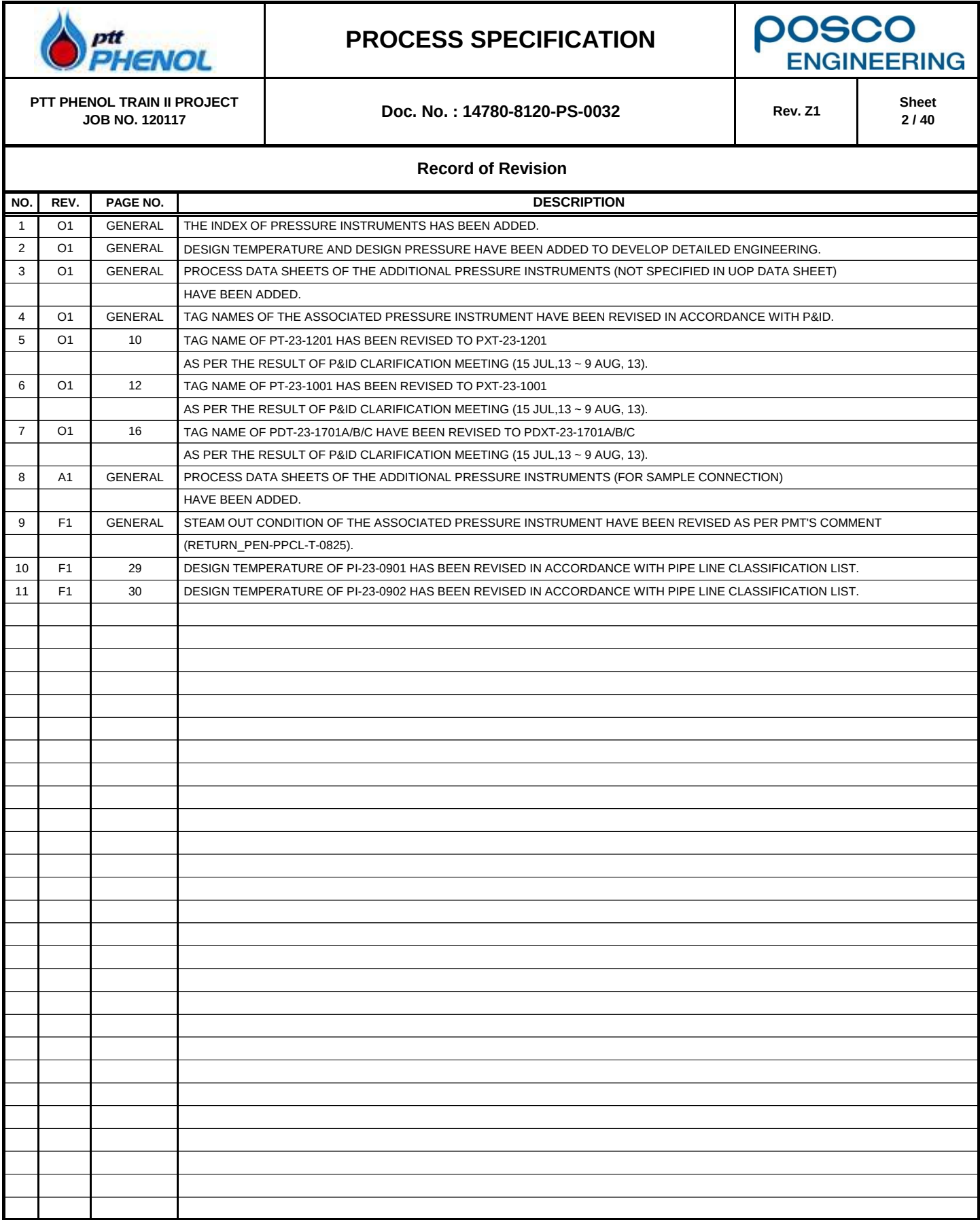
PRESSURE INSTRUMENT PROCESS DATA SHEET

(CDN UNIT)

DISCIPLINE : PROCESS
DOCUMENT NO. : 14780-8120-PS-0032
PROJECT NAME : PTT Phenol Train II Project
PROJECT NO. : 120117
OWNER : PTT Phenol Company Limited (PPCL)

Z1	MAY 10, 2016	AS BUILT	S.W.Park	J.S.Chung	-	S.W.Lee
F1	JUN 25, 2014	FINAL ISSUE	S.W.Park	J.S.Chung	-	S.W.Lee
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REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW	APPROVAL	
Z1	PROCESS	S.W.Park	J.S.Chung	-	S.W.Lee	
		MAY 10, 2016	MAY 10, 2016	-	MAY 10, 2016	

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PROJECT SPECIFICATION

SHEET 3 / 40

REV	DATE	BY	APVD	REV	DATE	BY	APVD

Rev

Concentration, Decomposition, and Neutralization Section

ITEM NUMBER	SERVICE	SHEET
INSTRUMENT GENERAL REQUIREMENT		6 - 7
GENERAL REQUIRMENT AND INFORMATION FOR PRESSURE INSTRUMENT		8
PT-23-0701	S3 Steam to Flash Column Vaporizer	9
PXT-23-1201	Process Water to Decomposer Feed	10
PT-23-0501	S1.5 Steam to Preflash Column Steam Heater	11
PT-23-1101	Cumene Flush Drum	11
PT-23-1202	Decomposer Feed Flush Drum	11
PT-23-0401	Preflash Column Top	12
PT-23-0402	Preflash Column Bottom	12
PT-23-0801	Flash Column Top	12
PT-23-0802	Flash Column Bottom	12
PXT-23-1001	Preflash and Flash Column Overhead Vapor to Flash Column Overhead Vapor Chiller	12
PT-23-1002	Preflash and Flash Column Overhead Vapor to Flash Column Overhead Vapor Chiller	12
PT-23-1401	Crude Product to Crude Product Cooler	13
PT-23-1301	Decomposer Drum	14
GENERAL REQUIRMENT AND INFORMATION FOR DIFFERENTIAL PRESSURE INSTRUMENT		15
PDT-23- 0301	Preflash Column Feed Filters dP	16
PDXT-23- 1701B	Decomposer Cooler dP	16
PDXT-23- 1701A	Decomposer Cooler dP	16
PDXT-23- 1701C	Decomposer Cooler dP	16
PDT-23- 1902	Direct Neutralization Static Mixers dP	17
PDT-23- 1401	Dehydrator	18
	Note for Differential Pressure Instruments	19 - 21
	Differential Pressure Instruments Sketch C	22
GENERAL REQUIREMENTS AND INFORMATION FOR PRESSURE GAUGE		23
PI-23-1504	Decomposer Acid Injection Pump A Discharge	24
PI-23-1505	Decomposer Acid Injection Pump B Discharge	24
PI-23-1601	Decomposer Acid Injection Pump C Discharge	24
PI-23-1602	Decomposer Acid Injection Pump D Discharge	24
PI-23-1603	Decomposer Acid Injection Pump E Discharge	24
PI-23-1604	Decomposer Acid Injection Pump F Discharge	24

Other specifications referenced in this specification:	UOP Standard Specification:
	UOP Standard Drawing:
	UOP Project Specification:



PROJECT SPECIFICATION

SHEET 4 / 40

PRESSURE INSTRUMENT

REV	DATE	BY	APVD	REV	DATE	BY	APVD

Rev

Concentration, Decomposition, and Neutralization Section

ITEM NUMBER	SERVICE	SHEET
PI-23-0302	Preflash Column Feed Filter A Inlet	25
PI-23-0303	Preflash Column Feed Filter B Inlet	25
PI-23-0304	Preflash Column Feed Filters Outlet	25
PI-23-0403	Preflash Column Top	25
PI-23-0404	Preflash Column Bottom	26
PI-23-0502	S1.5 Steam to Preflash Column Steam Heater	26
PI-23-0503	Preflash Column Steam Heater Condensate Pump A	26
PI-23-0504	Preflash Column Steam Heater Condensate Pump B	26
PI-23-0601	Concentration Cumene Quench Drum	27
PI-23-0702	S3 Steam to Flash Column Vaporizer	27
PI-23-0703	Sample Receiver Outlet	27
PI-23-0704	Flash Column Vaporizer Condensate Pump A	27
PI-23-0705	Flash Column Vaporizer Condensate Pump B	28
PI-23-0803	Flash Column Top	28
PI-23-0804	Flash Column Bottom	28
PI-23-0805	Reliable Cooling Water from Flash Column Bottoms Cooler	28
PI-23-0806	Reliable Cooling Water from Preflash and Flash Column Condenser	29
PI-23-0807	Preflash and Flash Columns Overhead Pump A	29
PI-23-0808	Preflash and Flash Columns Overhead Pump B	29
PI-23-0901	Flash Column Bottoms Pump A	29
PI-23-0902	Flash Column Bottoms Pump B	30
PI-23-1004	Preflash and Flash Column Overhead Vapor to Flash Column Overhead Vapor Chiller	30
PI-23-1005	Chilled Water from Flash Column Overhead Vapor Chiller	30
PI-23-1006	Chilled Water from Concentration Vacuum Producing Equipment	30
PI-23-1007	Nitrogen Purge to Concentration Vacuum Producing Equipment	31
PI-23-1102	Cumene Flush Drum	31
PI-23-1203	Decomposer Feed Flush Drum	31
PI-23-1204	Process Water to Decomposer Feed	31
PI-23-1302	Decomposer Drum	32
PI-23-1303	Decomposer Drum	32
PI-23-1304	Decomposer Product Pump A	32
PI-23-1305	Decomposer Product Pump B	32
PI-23-1402	Crude Product to Crude Product Cooler	33
PI-23-1403	Cooling Water from Crude Product Cooler	33
PI-23-1404	S4 Steam to Dehydrators	33
PI-23-1405	Dehydrator A Outlet	33
PI-23-1406	Dehydrator B Outlet	34
PI-23-1407	Dehydrators Inlet	34
PI-23-1702	Decomposer Circulation Pump A	34
PI-23-1703	Decomposer Circulation Pump B	34
PI-23-1704	Reliable Cooling Water from Decomposer Coolers	
PI-23-1801	Circulating Decomposer Liquid to Calorimeter No. 1 Inlet	35

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing:
UOP Project Specification:



PROJECT SPECIFICATION

SHEET 5 / 40

PRESSURE INSTRUMENT

REV	DATE	BY	APVD	REV	DATE	BY	APVD

Rev

Concentration, Decomposition, and Neutralization Section

ITEM NUMBER	SERVICE	SHEET
PI-23-1802	Circulating Decomposer Liquid from Calorimeter No. 1	35
PI-23-1803	Circulating Decomposer Liquid to Calorimeter No. 2 Inlet	35
PI-23-1804	Circulating Decomposer Liquid from Calorimeter No. 2	36
PI-23-1903	Direct Neutralization Static Mixers Inlet	36
PI-23-1904	Direct Neutralization Static Mixers Outlet	36
PI-23-1905	Neutralizing Agent Injection Pump A	36
PI-23-1906	Neutralizing Agent Injection Pump B	37
PI-23-2001	Acid Aromatics Knockout Drum	37
PI-23-2002	Acid Aromatics Sump Pump Outlet	37
PI-23-2003	Nitrogen Purge to Acid Aromatics Sump	37
PI-23-0813	SN-2301	38
PI-23-0907	SN-2303	38
PI-23-0706	SN-2310	38
PI-23-2101	AD Funnel	39
PI-23-2301	CHD Funnel	39
	Note for Pressure Gauges	40

Other specifications referenced in this specification:	UOP Standard Specification:
	UOP Standard Drawing:
	UOP Project Specification:

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Indication

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INSTRUMENT GENERAL REQUIREMENTS
PROJECT SPECIFICATION

 963766 — **601**

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Phenol Concentration, Decomposition, and Neutralization Section

This specification shall be supplementary to all Project Specifications, 602 - 699, issued under this project number. Unless specified otherwise on the Project Specification, the following shall apply:

1 Climatic Ambient Temperature Range

The climatic ambient temperature range for field equipment design is 15 to 40°C.

2 Electrical Classification

The area electrical classification shall be established by the Licensee/Contractor in accordance with local electrical codes. The Contractor shall advise all equipment vendors of the area electrical classification.

3 Flanges

3.1 Carbon steel flanges are based on Material Group 1.1 of ASME B16.5 and ASME B16.47. Flange classes may differ when instrument flanges and the mating flanges on piping or equipment are specified with dissimilar materials. The Contractor shall verify that all mating flanges are properly matched with the specified instrument flanges.

3.2 Flange classes are specified in accordance with ASME B16.5 or ASME B16.47. Flange classes listed in the UOP Project Specifications are based upon design pressure and temperature conditions only and do not account for other loads. The final design of all flanges shall account for gasket seating and external loads. Differential thermal expansion, including joints between dissimilar materials, and transient thermal conditions such as start-up/shutdown and operational upset shall be accommodated.

3.3 Flanges intended for use with spiral wound gaskets shall have a flange surface finish of 125 microinch Ra minimum to 250 microinch Ra maximum. Flanges intended for use with other gaskets shall have a flange finish within the optimal range for the specified gasket. Finishes shall be judged by visual comparison with surface finish roughness standards conforming to ASME B46.1. Flange finishes shall be protected from damage during fabrication, heat treatment, shipping, storage, and installation.

4 Tropical Environment

All equipment shall be tropicalized to eliminate mildew, fungi, and other detrimental effects of a tropical environment. Packaging shall be suitable for shipment and storage under tropical conditions. Packaging and "tropicalization" subject to approval by the licensee.

5 Marine Environment

All field equipment shall be suitable for operation in a corrosive, salt laden, marine atmosphere.

6 Tropical and Marine Environment

Transmitters and all equipment/instruments containing electrical parts shall be installed under rain shield.

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INSTRUMENT GENERAL REQUIREMENTS

PROJECT SPECIFICATION

963766 — 601

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Phenol Concentration, Decomposition, and Neutralization Section

7 Fieldbus Transmitters

This specification is based on conventional (4 – 20 mA output) transmitters. UOP does not require the use of fieldbus transmitters and, if installed incorrectly, the fieldbus system can be detrimental to the control and operation of the unit. The implementation of these fieldbus transmitters is the Contractor's responsibility.

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A Honeywell Company



PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 603-G

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION

1 Standard Drawings Referenced

UOP Standard Drawings as indicated below are part of this specification.

X	6-102		6-122 + 6-123		6-146		6-155			
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2 Sensor Range

The Sensing Element Lower Range Limit and Upper Range Limit are to be used as a guide only. Transmitters that have greater range capabilities are acceptable.

3 Output Indicator

When an Output Indicator is specified, an integrally mounted indicator is acceptable, provided the indicator is readable from the associated control valve assembly (or other valve as denoted on the P&ID). If the piping layout is such that an integrally mounted indicator is not readable from the associated valve, then a separate indicator suitable for remote mounting shall be supplied.

4 Process Temperature

The temperature shown is the *process* temperature and is not normally the operating temperature of the transmitter.

Other specifications
referenced in this
specification:

UOP Standard Specification: 6-18
UOP Standard Drawing: See item 1 above
UOP Project Specification: 601, this project

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  PRESSURE INSTRUMENTS					PROJECT SPECIFICATION 963766 — 603 <table><tr><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td></tr><tr><td>0</td><td>25JUN12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>				REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	25JUN12	MVD	MW																												
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0	25JUN12	MVD	MW																																													
Item Number: 1 Quantity: 1 Data Page: 1 of 1																																																
BODY	Body/Flange Material: Cadmium plated CS or 316ss			REMOTE SEAL	Process Conn Size: Type: Class:																																											
	Static Pressure Rating: 105 kg/cm2(g) ‡				Flange Facing: Flange Face Finish:																																											
	Process Connections Size: 1/2 inch				Seal Type:																																											
	Process Connections Material: Cadmium plated CS or 316ss				Fill Fluid:																																											
	Drains/Vents Material: 316 stainless steel				Diaphragm Material:																																											
	Gasket/O-Ring Material: Man. Recommendation				Mounting Flange Material:																																											
	Gasket Type: Manufacturer's Recommendation				Non-Wetted Housing Material:																																											
	Bolting/Nut Material: Manufacturer's Recommendation				Capillary Length: Armor Material:																																											
					Diaphragm Extension Length:																																											
SENSING ELEMENT	Diaphragm/Wetted Material: 316 stainless steel			INDICATOR / CONTROLLER	Function:																																											
	Non-Wetted Material: Manufacturer's Recommendation				Controller Type:																																											
	Zero Elevation/Suppression: Provide				Controller Action:																																											
	Sensor Lower Range Limit: 1.20 kg/cm2 gauge				Controller Output:																																											
	Sensor Upper Range Limit: 7.0 kg/cm2 gauge				Manufacturer:																																											
Fill Fluid: Silicone																																																
TRANSMITTER	Transmitter Output: 4 - 20 mA			ACCESSORIES	Manifold Type:																																											
	Adjustable Dampening:				Manifold Material:																																											
	Housing Material: Manufacturer's Recommendation				Manifold Manufacturer:																																											
	Manufacturer: Honeywell, Rosemount (or equal)				Output Indicator: 0 - 100 linear																																											
			Air Filter/Regulator Quantity:																																													
REMARKS	‡ Static pressure rating shall be a minimum of 150 percent of the Sensor Upper Range Limit at the operating temperature.																																															
	DP : Design Pressure																																															
	DT : Design Temperature																																															
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments																																									
	PT-23-0701	S3 Steam to Flash Column Vaporizer	0 to 3.5 kg/cm2(g)	135 °C	1.8 kg/cm2(g)	6-102	DP : 10.5 kg/cm2(g), F.V / DT : 195 degC																																									
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																



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PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 603

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 2		Quantity: 1		Data Page: 1 of 1			
BODY	Body/Flange Material: Cadmium plated CS or 316ss			REMOTE SEAL	Process Conn Size: _____ Type: _____ Class: _____		
	Static Pressure Rating: 105 kg/cm2(g) ‡				Flange Facing: _____ Flange Face Finish: _____		
	Process Connections Size: 1/2 inch				Seal Type: _____		
	Process Connections Material: Cadmium plated CS or 316ss				Fill Fluid: _____		
	Drains/Vents Material: 316 stainless steel				Diaphragm Material: _____		
	Gasket/O-Ring Material: Man. Recommendation				Mounting Flange Material: _____		
	Gasket Type: Manufacturer's Recommendation				Non-Wetted Housing Material: _____		
	Bolting/Nut Material: Manufacturer's Recommendation				Capillary Length: _____ Armor Material: _____		
	Diaphragm/Wetted Material: 316 stainless steel				Diaphragm Extension Length: _____		
	Non-Wetted Material: Manufacturer's Recommendation				INDICATOR / CONTROLLER	Function: _____	
Zero Elevation/Suppression: Provide			Controller Type: _____				
Sensor Lower Range Limit: 3.52 kg/cm2 gauge			Controller Action: _____				
Sensor Upper Range Limit: 21.0 kg/cm2 gauge			Controller Output: _____				
Fill Fluid: Silicone			Manufacturer: _____				
TRANSMITTER	Transmitter Output: 4 - 20 mA			ACCESSORIES	Manifold Type: _____		
	Adjustable Dampening: _____				Manifold Material: _____		
	Housing Material: Manufacturer's Recommendation				Manifold Manufacturer: _____		
	Manufacturer: Honeywell, Rosemount (or equal)				Output Indicator: 0 - 100 linear		
				Air Filter/Regulator Quantity: _____			
REMARKS	‡ Static pressure rating shall be a minimum of 150 percent of the Sensor Upper Range Limit at the operating temperature.						
	DP : Design Pressure						
	DT : Design Temperature						
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments
	PXT-23-1201	Process Water to Decomposer Feed	0 to 8 kg/cm2(g)	50 °C	5.4 kg/cm2(g)	6-102	DP : 9.716 kg/cm2(g) / DT : 120 degC
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.							

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  PRESSURE INSTRUMENTS					PROJECT SPECIFICATION 963766 — 603 <table><tr><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td></tr><tr><td>0</td><td>25JUN12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>				REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	25JUN12	MVD	MW																												
REV	DATE	BY	APVD	REV	DATE	BY	APVD																																									
0	25JUN12	MVD	MW																																													
Item Number: 3 Quantity: 3 Data Page: 1 of 1																																																
BODY	Body/Flange Material: Cadmium plated CS or 316ss			REMOTE SEAL	Process Conn Size: Type: Class:																																											
	Static Pressure Rating: 105 kg/cm2(g) ‡				Flange Facing: Flange Face Finish:																																											
	Process Connections Size: 1/2 inch				Seal Type:																																											
	Process Connections Material: Cadmium plated CS or 316ss				Fill Fluid:																																											
	Drains/Vents Material: 316 stainless steel				Diaphragm Material:																																											
	Gasket/O-Ring Material: Man. Recommendation				Mounting Flange Material:																																											
	Gasket Type: Manufacturer's Recommendation				Non-Wetted Housing Material:																																											
	Bolting/Nut Material: Manufacturer's Recommendation				Capillary Length: Armor Material:																																											
					Diaphragm Extension Length:																																											
SENSING ELEMENT	Diaphragm/Wetted Material: 316 stainless steel			INDICATOR / CONTROLLER	Function:																																											
	Non-Wetted Material: Manufacturer's Recommendation				Controller Type:																																											
	Zero Elevation/Suppression: Provide				Controller Action:																																											
	Sensor Lower Range Limit: 300 mBar gauge				Controller Output:																																											
	Sensor Upper Range Limit: 1850 mBar gauge				Manufacturer:																																											
	Fill Fluid: Silicone																																															
TRANSMITTER	Transmitter Output: 4 - 20 mA			ACCESSORIES	Manifold Type:																																											
	Adjustable Dampening:				Manifold Material:																																											
	Housing Material: Manufacturer's Recommendation				Manifold Manufacturer:																																											
	Manufacturer: Honeywell, Rosemount (or equal)				Output Indicator: 0 - 100 linear																																											
					Air Filter/Regulator Quantity:																																											
REMARKS	‡ Static pressure rating shall be a minimum of 150 percent of the Sensor Upper Range Limit at the operating temperature.																																															
	DP : Design Pressure																																															
	DT : Design Temperature																																															
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments																																									
	PT-23-0501	S1.5 Steam to Preflash Column Steam Heater	0 to 1.5 kg/cm2(g)	121 °C	0.7 kg/cm2(g)	6-102	DP : 7.0 kg/cm2(g), F.V / DT : 195 degC																																									
	PT-23-1101	Cumene Flush Drum	0 to 1.5 kg/cm2(g)	46 °C	0.7 kg/cm2(g)	6-102	DP : 10.0 kg/cm2(g) / DT : 120 degC																																									
	PT-23-1202	Decomposer Feed Flush Drum	0 to 1.5 kg/cm2(g)	46 °C	0.7 kg/cm2(g)	6-102	DP : 10.0 kg/cm2(g) / DT : 120 degC																																									
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																



PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 603

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 4		Quantity: 6		Data Page: 1 of 1			
BODY	Body/Flange Material: 316 stainless steel ‡			REMOTE SEAL	Process Conn Size: Type: Class:		
	Static Pressure Rating: 10.5 kg/cm2(a)				Flange Facing: Flange Face Finish:		
	Process Connections Size: 1/2 inch				Seal Type:		
	Process Connections Material: 316 stainless steel ‡				Fill Fluid:		
	Drains/Vents Material: 316 stainless steel ‡				Diaphragm Material:		
	Gasket/O-Ring Material: Man. Recommendation				Mounting Flange Material:		
	Gasket Type: Manufacturer's Recommendation				Non-Wetted Housing Material:		
	Bolting/Nut Material: Manufacturer's Recommendation				Capillary Length: Armor Material:		
					Diaphragm Extension Length:		
	SENSING ELEMENT	Diaphragm/Wetted Material: 316 stainless steel ‡			INDICATOR / CONTROLLER	Function:	
Non-Wetted Material: Manufacturer's Recommendation			Controller Type:				
Zero Elevation/Suppression: Provide			Controller Action:				
Sensor Lower Range Limit: 9 mm Hg ABSOLUTE			Controller Output:				
Sensor Upper Range Limit: 260 mm Hg ABSOLUTE			Manufacturer:				
Fill Fluid: Silicone							
TRANSMITTER	Transmitter Output: 4 - 20 mA			ACCESSORIES	Manifold Type:		
	Adjustable Dampening:				Manifold Material:		
	Housing Material: Manufacturer's Recommendation				Manifold Manufacturer:		
	Manufacturer: Rosemount, Foxboro (or equal)				Output Indicator: 0 - 100 linear		
				Air Filter/Regulator Quantity:			
REMARKS	ABSOLUTE pressure transmitter						
	‡ All transmitter parts in contact with the process shall be 316 stainless steel.						
	DP : Design Pressure						
	DT : Design Temperature						
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments
	PT-23-0401	Preflash Column Top	0 to 75 mm Hg(a)	52 °C	15 mm Hg(a)	6-102	Impulse piping and fittings, shall be 316 stainless steel. DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC
	PT-23-0402	Preflash Column Bottom	0 to 75 mm Hg(a)	70 °C	15 mm Hg(a)	6-102	Impulse piping and fittings, shall be 316 stainless steel. DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC
	PT-23-0801	Flash Column Top	0 to 75 mm Hg(a)	51 °C	15 mm Hg(a)	6-102	Impulse piping and fittings, shall be 316 stainless steel. DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC
	PT-23-0802	Flash Column Bottom	0 to 150 mm Hg(a)	95 °C	30 mm Hg(a)	6-102	Impulse piping and fittings, shall be 316 stainless steel. DP : 3.5 kg/cm2(g), F.V / DT : 250, 195 degC
	PXT-23-1001	Preflash and Flash Column Overhead Vapor to Flash Column Overhead Vapor Chiller	0 to 50 mm Hg(a)	38 °C	8 mm Hg(a)	6-102	Impulse piping and fittings, shall be 316 stainless steel. DP : 4.5 kg/cm2(g), F.V / DT : 120 degC
	PT-23-1002	Preflash and Flash Column Overhead Vapor to Flash Column Overhead Vapor Chiller	0 to 40 mm Hg(a)	38 °C	8 mm Hg(a)	6-102	Impulse piping and fittings, shall be 316 stainless steel. DP : 4.5 kg/cm2(g), F.V / DT : 120 degC
	ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.						

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  PRESSURE INSTRUMENTS					PROJECT SPECIFICATION 963766 — 603 <table><tr><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td><td>REV</td><td>DATE</td><td>BY</td><td>APVD</td></tr><tr><td>0</td><td>25JUN12</td><td>MVD</td><td>MW</td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>				REV	DATE	BY	APVD	REV	DATE	BY	APVD	0	25JUN12	MVD	MW																												
REV	DATE	BY	APVD	REV	DATE	BY	APVD																																									
0	25JUN12	MVD	MW																																													
Item Number: 5 Quantity: 1 Data Page: 1 of 1																																																
BODY	Body/Flange Material: 316 stainless steel †				REMOTE SEAL	Process Conn Size: Type: Class:																																										
	Static Pressure Rating: 105 kg/cm2(g) ‡					Flange Facing: Flange Face Finish:																																										
	Process Connections Size: 1/2 inch					Seal Type:																																										
	Process Connections Material: 316 stainless steel †					Fill Fluid:																																										
	Drains/Vents Material: 316 stainless steel †					Diaphragm Material:																																										
	Gasket/O-Ring Material: Man. Recommendation					Mounting Flange Material:																																										
	Gasket Type: Manufacturer's Recommendation					Non-Wetted Housing Material:																																										
	Bolting/Nut Material: Manufacturer's Recommendation					Capillary Length: Armor Material:																																										
						Diaphragm Extension Length:																																										
SENSING ELEMENT	Diaphragm/Wetted Material: 316 stainless steel †				INDICATOR / CONTROLLER	Function:																																										
	Non-Wetted Material: Manufacturer's Recommendation					Controller Type:																																										
	Zero Elevation/Suppression: Provide					Controller Action:																																										
	Sensor Lower Range Limit: 1.20 kg/cm2 gauge					Controller Output:																																										
	Sensor Upper Range Limit: 7.0 kg/cm2 gauge					Manufacturer:																																										
Fill Fluid: Silicone																																																
TRANSMITTER	Transmitter Output: 4 - 20 mA				ACCESSORIES	Manifold Type:																																										
	Adjustable Dampening:					Manifold Material:																																										
	Housing Material: Manufacturer's Recommendation					Manifold Manufacturer:																																										
	Manufacturer: Honeywell, Rosemount (or equal)					Output Indicator: 0 - 100 linear																																										
				Air Filter/Regulator Quantity:																																												
REMARKS	‡ Static pressure rating shall be a minimum of 150 percent of the Sensor Upper Range Limit at the operating temperature.																																															
	† All transmitter parts in contact with the process shall be 316 stainless steel.																																															
	DP : Design Pressure																																															
	DT : Design Temperature																																															
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments																																									
	PT-23-1401	Crude Product to Crude Product Cooler	0 to 7 kg/cm2(g)	140 °C	5.1 kg/cm2(g)	6-102	Impulse piping and fittings, shall be 316 stainless steel.																																									
							DP : 21.0 kg/cm2(g), F.V / DT : 250 degC																																									
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																

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REV	DATE	BY	APVD	REV	DATE	BY	APVD																																									
0	25JUN12	MVD	MW																																													
Item Number: 6 Quantity: 1 Data Page: 1 of 1																																																
BODY	Body/Flange Material: 316 stainless steel †			REMOTE SEAL	Process Conn Size: Type: Class:																																											
	Static Pressure Rating: 105 kg/cm2(g) ‡				Flange Facing: Flange Face Finish:																																											
	Process Connections Size: 1/2 inch				Seal Type:																																											
	Process Connections Material: 316 stainless steel †				Fill Fluid:																																											
	Drains/Vents Material: 316 stainless steel †				Diaphragm Material:																																											
	Gasket/O-Ring Material: Man. Recommendation				Mounting Flange Material:																																											
	Gasket Type: Manufacturer's Recommendation				Non-Wetted Housing Material:																																											
	Bolting/Nut Material: Manufacturer's Recommendation				Capillary Length: Armor Material:																																											
					Diaphragm Extension Length:																																											
TRANSMITTER SENSING ELEMENT	Diaphragm/Wetted Material: 316 stainless steel †			INDICATOR / CONTROLLER	Function:																																											
	Non-Wetted Material: Manufacturer's Recommendation				Controller Type:																																											
	Zero Elevation/Suppression: Provide				Controller Action:																																											
	Sensor Lower Range Limit: 300 mBar gauge				Controller Output:																																											
	Sensor Upper Range Limit: 1850 mBar gauge				Manufacturer:																																											
Fill Fluid: Silicone																																																
TRANSMITTER	Transmitter Output: 4 - 20 mA			ACCESSORIES	Manifold Type:																																											
	Adjustable Dampening:				Manifold Material:																																											
	Housing Material: Manufacturer's Recommendation				Manifold Manufacturer:																																											
	Manufacturer: Honeywell, Rosemount (or equal)				Output Indicator: 0 - 100 linear																																											
			Air Filter/Regulator Quantity:																																													
REMARKS	‡ Static pressure rating shall be a minimum of 150 percent of the Sensor Upper Range Limit at the operating temperature.																																															
	† All transmitter parts in contact with the process shall be 316 stainless steel.																																															
	DP : Design Pressure																																															
	DT : Design Temperature																																															
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments																																									
	PT-23-1301	Decomposer Drum	0 to 1.5 kg/cm2(g)	60 °C	0.7 kg/cm2(g)	6-102	Impulse piping and fittings, shall be 316 stainless steel.																																									
							DP : 11.0 kg/cm2(g), F.V / DT : 250 degC																																									
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																

Revision
Indication

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ENGINEERING

DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-G

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION

1 Standard Drawings Referenced

UOP Standard Drawings as indicated below are part of this specification.

6-115	X	6-116	6-120	6-122 + 6-123	6-139	6-140
6-142	X	6-144	6-147	6-148	6-159	6-161
						6-162

2 Sensor Range

The Sensing Element Lower Range Limit and Upper Range Limit are to be used as a guide only. Transmitters that have greater range capabilities are acceptable.

3 Output Indicator

When an Output Indicator is specified, an integrally mounted indicator is acceptable, provided the indicator is readable from the associated control valve assembly (or other valve as denoted on the P&ID). If the piping layout is such that an integrally mounted indicator is not readable from the associated valve, then a separate indicator suitable for remote mounting shall be supplied.

4 Process Temperature

The temperature shown is the *process* temperature and is not normally the operating temperature of the transmitter.

Other specifications
referenced in this
specification:

UOP Standard Specification: 6-18
UOP Standard Drawing: See item 1 above
UOP Project Specification: 601, this project



DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

	Item Number: 5	Quantity: 4	Data Page: 1 of 1				
BODY	Body/Flange Material: 316 stainless steel ‡				HIGH SIDE REMOTE SEAL	Process Conn Size: Type: Class:	
	Static Pressure Rating: 105 kg/cm2(g)					Flange Facing: Flange Face Finish:	
	Process Connections Size: 1/2 inch					Seal Type:	
	Process Connections Material: 316 stainless steel ‡					Fill Fluid:	
	Drains/Vents Material: 316 stainless steel ‡					Diaphragm Material:	
	Gasket/O-Ring Material: Man. Recommendation					Mounting Flange Material:	
	Gasket Type: Manufacturer's Recommendation					Non-Wetted Housing Material:	
	Bolting/Nut Material: Manufacturer's Recommendation					Capillary Length: Armor Material:	
TRANSMITTER SENSING ELEMENT	Diaphragm/Wetted Material: 316 stainless steel ‡				LOW SIDE REMOTE SEAL	Diaphragm Extension Length:	
	Non-Wetted Material: Manufacturer's Recommendation					Process Conn Size: Type: Class:	
	Zero Elevation/Suppression: Provide					Flange Facing: Flange Face Finish:	
	Sensor Lower Range Limit: 1.20 kg/cm2					Seal Type:	
	Sensor Upper Range Limit: 7.0 kg/cm2					Fill Fluid:	
INDICATOR / CONTROLLER	Fill Fluid: Silicone				ACCESSORIES	Diaphragm Material:	
	Transmitter Output: 4 - 20 mA					Mounting Flange Material:	
	Adjustable Dampening:					Non-Wetted Housing Material:	
	Housing Material: Manufacturer's Recommendation					Capillary Length: Armor Material:	
REMARKS	Manufacturer: Honeywell, Rosemount (or equal)					Diaphragm Extension Length:	
	Function:					Manifold Type:	
	Controller Type:					Manifold Material:	
	Controller Action:					Manifold Manufacturer:	
DISTRIBUTION	Controller Output:					Output Indicator: See Remark §	
	Manufacturer:					Air Filter/Regulator Quantity:	
	§ 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential.						
	‡ All transmitter parts in contact with the process shall be 316 stainless steel.						
DP : Design Pressure							
DT : Design Temperature							
DISTRIBUTION	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments
	PDT-23-0301	Preflash Column Feed Filters dP	0 to 3 kg/cm2	82 °C	8.5 kg/cm2(g)	6-144 Detail A	Impulse piping and fittings, shall be 316 stainless steel. DP : 11.5 kg/cm2(g), F.V / DT : 120 degC
	PDXT-23-1701B	Decomposer Cooler dP	0 to 2 kg/cm2	79 °C	3.7 kg/cm2(g)	6-144 Detail A	Impulse piping and fittings, shall be 316 stainless steel. DP : 16.5 kg/cm2(g), F.V / DT : 250 degC
	PDXT-23-1701A	Decomposer Cooler dP	0 to 2 kg/cm2	79 °C	3.7 kg/cm2(g)	6-144 Detail A	Impulse piping and fittings, shall be 316 stainless steel. DP : 16.5 kg/cm2(g), F.V / DT : 250 degC
	PDXT-23-1701C	Decomposer Cooler dP	0 to 2 kg/cm2	79 °C	3.7 kg/cm2(g)	6-144 Detail A	Impulse piping and fittings, shall be 316 stainless steel. DP : 16.5 kg/cm2(g), F.V / DT : 250 degC

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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REV	DATE	BY	APVD	REV	DATE	BY	APVD																																									
0	25JUN12	MVD	MW																																													
Item Number: 7 Quantity: 4 Data Page: 1 of 1																																																
BODY	Body/Flange Material: 316 stainless steel ‡				HIGH SIDE REMOTE SEAL	Process Conn Size: Type: Class:																																										
	Static Pressure Rating: 105 kg/cm2(g)					Flange Facing: Flange Face Finish:																																										
	Process Connections Size: 1/2 inch					Seal Type:																																										
	Process Connections Material: 316 stainless steel ‡					Fill Fluid:																																										
	Drains/Vents Material: 316 stainless steel ‡					Diaphragm Material:																																										
	Gasket/O-Ring Material: Man. Recommendation					Mounting Flange Material:																																										
	Gasket Type: Manufacturer's Recommendation					Non-Wetted Housing Material:																																										
	Bolting/Nut Material: Manufacturer's Recommendation					Capillary Length: Armor Material:																																										
TRANSMITTER SENSING ELEMENT	Diaphragm/Wetted Material: 316 stainless steel ‡				LOW SIDE REMOTE SEAL	Diaphragm Extension Length:																																										
	Non-Wetted Material: Manufacturer's Recommendation					Process Conn Size: Type: Class:																																										
	Zero Elevation/Suppression: Provide					Flange Facing: Flange Face Finish:																																										
	Sensor Lower Range Limit: 300 mBar					Seal Type:																																										
	Sensor Upper Range Limit: 1860 mBar					Fill Fluid:																																										
INDICATOR / CONTROLLER	Fill Fluid: Silicone				ACCESSORIES	Diaphragm Material:																																										
	Transmitter Output: 4 - 20 mA					Mounting Flange Material:																																										
	Adjustable Dampening:					Non-Wetted Housing Material:																																										
	Housing Material: Manufacturer's Recommendation					Capillary Length: Armor Material:																																										
REMARKS	Manufacturer: Honeywell, Rosemount (or equal)					Diaphragm Extension Length:																																										
	Function:					Manifold Type:																																										
	Controller Type:					Manifold Material:																																										
	Controller Action:					Manifold Manufacturer:																																										
DISTRIBUTION	Controller Output:					Output Indicator: See Remark §																																										
	Manufacturer:					Air Filter/Regulator Quantity:																																										
	§ 0 - 10 Sq. Rt. for flow; 0 - 100 linear for level and pressure differential.																																															
	‡ All transmitter parts in contact with the process shall be 316 stainless steel.																																															
	DP : Design Pressure																																															
	DT : Design Temperature																																															
	Tag Number	Service	Calibrated Range	Process Temperature	Process Pressure	Installation	Comments																																									
	PDT-23-1902	Direct Neutralization Static Mixers dP	0 to 1.5 kg/cm2	43 °C	4.3 kg/cm2(g)	6-144 Detail A	Impulse piping and fittings, shall be 316 stainless steel.																																									
							DP : 21.0 kg/cm2(g), F.V / DT : 250 degC																																									
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.																																																



DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Item Number: 8 Quantity: 1 Data Page: 1 of 1							
BODY	Body/Flange Material: Cadmium plated CS or 316ss Static Pressure Rating: 105 kg/cm2(g) Process Connections Size: 1/2 inch Process Connections Material: Cadmium plated CS or 316ss Drains/Vents Material: 316 stainless steel Gasket/O-Ring Material: Man. Recommendation Gasket Type: Manufacturer's Recommendation Bolting/Nut Material: Manufacturer's Recommendation	HIGH SIDE REMOTE SEAL	Process Conn Size: 3 inch Type: Flange Class: 300 Flange Facing: Raised Face Flange Face Finish: Seal Type: Wafer Fill Fluid: Silicone § Diaphragm Material: 316 stainless steel Mounting Flange Material: Non-Wetted Housing Material: Mfr. Standard Capillary Length: ± Armor Material: Mfr. Std. Diaphragm Extension Length:				
	Diaphragm/Wetted Material: 316 stainless steel Non-Wetted Material: Manufacturer's Recommendation Zero Elevation/Suppression: Provide Sensor Lower Range Limit: 300 mBar Sensor Upper Range Limit: 1860 mBar Fill Fluid: Silicone		LOW SIDE REMOTE SEAL	Process Conn Size: 3 inch Type: Flange Class: 300 Flange Facing: Raised Face Flange Face Finish: Seal Type: Wafer Fill Fluid: Silicone § Diaphragm Material: 316 stainless steel Mounting Flange Material: Non-Wetted Housing Material: Mfr. Standard Capillary Length: ± Armor Material: Mfr. Std. Diaphragm Extension Length:			
	Transmitter Output: 4 - 20 mA Adjustable Dampening: Housing Material: Manufacturer's Recommendation Manufacturer: Honeywell, Rosemount (or equal)			ACCESSORIES	Manifold Type: Manifold Material: Manifold Manufacturer: Output Indicator: 0 - 100 Linear Air Filter/Regulator Quantity:		
	Function: Controller Type: Controller Action: Controller Output: Manufacturer:						
REMARKS § Capillary fill shall be suitable for the maximum operating temperature indicated. ‡ The capillary length shall be of sufficient length on each leg for installation of the transmitter between the taps. The capillary tubing shall be protected from direct contact with sunlight to minimize measurement errors caused by temperature differences between high and low capillaries. DP : Design Pressure DT : Design Temperature							
DISTRIBUTION	Tag Number	Service	Calibrated Range		Process Temperature	Process Pressure	Installation
	PDT-23-1401	Dehydrator	0 to 1.4 kg/cm2	135 °C	6.2 kg/cm2(g)	Sketch "C"	Capillary fill max temperature = 250°C Note 3 DP : 21.0 kg/cm2(g), F.V / DT : 250 degC
ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.							

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-N

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Notes:

- Four transmitters are piped from one orifice flange. The transmitters shall be installed in accordance with UOP Standard Drawing 6-116 and shall have individual isolation.
- The transmitter differential range specified is based upon a level range of ~~1600~~ 1200 mm and a liquid specific gravity of 0.916. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- Impulse piping and fittings shall be 316 stainless steel.
- The transmitter differential range specified is based upon a level range of 1200 mm and a liquid specific gravity of 0.916. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.947. Fill the reference leg with Water and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 400 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.931. Fill the reference leg with Water and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 2400 mm and a liquid specific gravity of 0.935. Provide zero elevation as required to compensate for the static head of capillary fill fluid between the upper and lower level connections and the transmitter. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
- The transmitter differential range specified is based upon a level range of 4200 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.

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Indication

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-N

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Notes:

10. The transmitter differential range specified is based upon a level range of 1400 mm and a liquid specific gravity of 0.841. Fill the reference leg with Cumene and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
11. The transmitter differential range specified is based upon a level range of 400 mm and a liquid specific gravity of 0.920. Fill the reference leg with Decomposer Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
12. The transmitter differential range specified is based upon a level range of 800 mm and a liquid specific gravity of 0.920. Fill the reference leg with Decomposer Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
13. Provide a first order lag signal filtering function, with adjustable time constant, for reduction of process signal noise. The signal filtering may be accomplished either by analog or digital means, at the transmitter. Alternatively, the signal filtering can be performed at the DCS. The time constant shall initially be set at the minimum value.
14. A horizontal long radius elbow in the regular piping shall be used for this meter. The pressure taps shall be located at an angle of 45 degrees from the inlet weld seam, diametrically opposite one another, at the extreme inside and outside of the bend. Provide 25 diameters of straight run pipe upstream of the long radius elbow and 10 diameters downstream. The meter manifold and meter installation shall otherwise conform to UOP Standard Drawing 6-116.
15. Two transmitters are piped from one orifice flange. The transmitters shall be installed in accordance with UOP Standard Drawing 6-116 and shall have individual isolation.
16. The transmitter differential range specified is based upon a level range of 2100 mm and a liquid specific gravity of 0.800. Fill the reference leg with Acid Aromatics Knockout Drum liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
17. The transmitter differential range specified is based upon a level range of 2500 mm and a liquid specific gravity of 0.848. Fill the reference leg with Preflash and Flash Columns Condenser liquid and provide zero elevation as required to compensate for the reference head. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.

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DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-N

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				
1	28AUG12	MVD	MW				

Notes:

18. The transmitter differential range specified is based upon a level range of 1000 mm and a liquid specific gravity of 1.716. Provide zero elevation as required to compensate for the static head of capillary fill fluid. ~~between the upper and lower level connections and the transmitter.~~ The low pressure tap shall be vented to the atmosphere and protected from rain, wind and insects. Consideration shall be given to elevation/suppression requirements if a different vendor is selected.
19. Contractor to confirm the calibration range.



posco
ENGINEERING

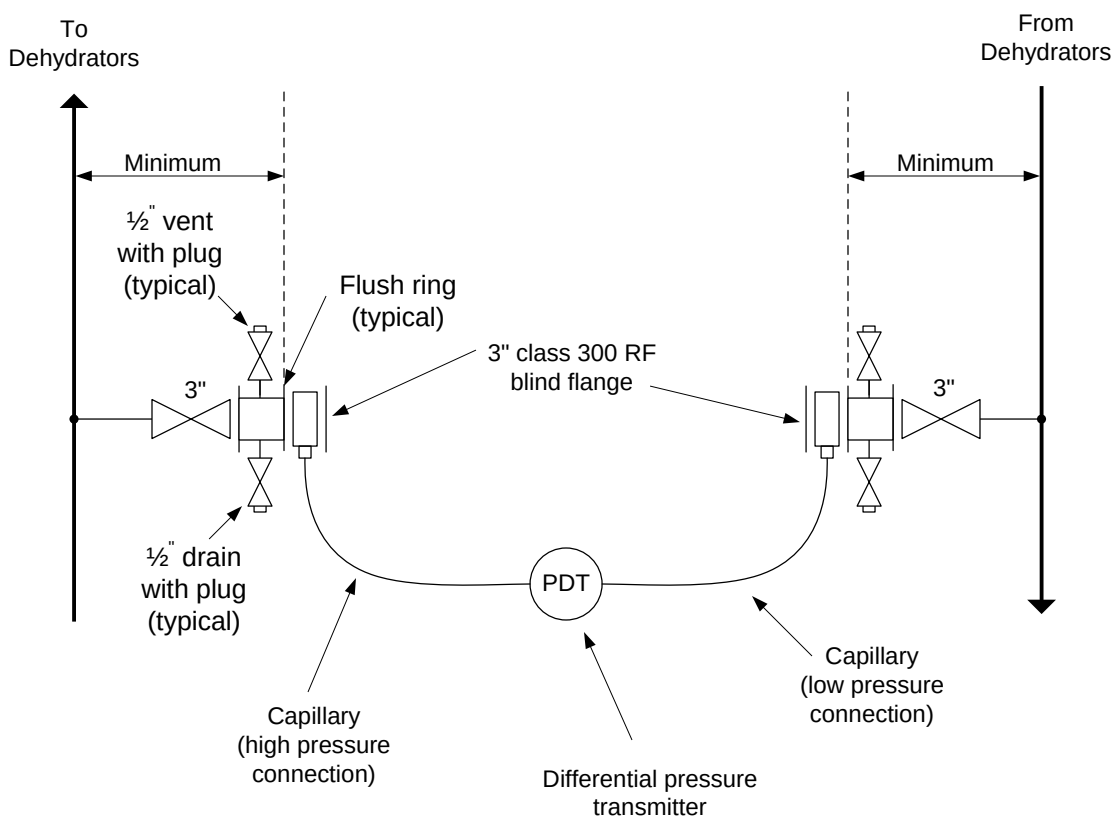
DIFFERENTIAL PRESSURE INSTRUMENTS

PROJECT SPECIFICATION

963766 — 604-D

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Sketch " C "



Notes:

1. Piping shall be in accordance with Project Specification 801, this project, and UOP Standard Specification 8-11.
2. Pipe and nipples between fittings shall be minimum length.
3. Locate nozzles on side of vertical or horizontal line.
4. Locate transmitter below lower tap.
5. All valves are gate valves.

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PRESSURE GAUGES

PROJECT SPECIFICATION

963766 — 602-G

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GENERAL REQUIREMENTS AND INFORMATION

1 Standard Drawings Referenced

UOP Standard Drawings as indicated below are part of this specification.

X	6-101										
---	-------	--	--	--	--	--	--	--	--	--	--

2 Gauge Accuracy

Guaranteed gauge accuracy shall be 1% of the scale range.

3 Overrange

Pressure elements shall be capable of withstanding intermittent over-ranging to 1.3 times the maximum scale reading without shifting calibration more than 1% of the scale reading.

4 Bourdon Tube

Bourdon tubes shall be welded to socket and tip and stress relieved as required.

5 Movement

Movement for all gauges shall be stainless steel or anodized aluminum and nylon.

6 Case

Cases shall be vapor tight and weatherproof.

7 Dial

Dials shall be white with black numerals and markings.

8 Calibration

All gauges shall be equipped with screw driver slot type adjustment for calibration purposes.

9 Vent Line Block Valve

Pressure gauge vent lines in UOP Standard Drawing 6-101 shall be provided with globe valves before the screwed caps.

Other specifications
referenced in this
specification:

UOP Standard Specification:
UOP Standard Drawing: See item 1 above
UOP Project Specification: 601, this project





PRESSURE GAUGES

PROJECT SPECIFICATION							
963766 —				602			
REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

GAUGE	Item Number: 1			Data Page:1 of 1		
	Type: Direct Reading Dial Diameter: 4-1/2" (100 mm) minimum Process Connection: 1/2 inch NPT Connection Location: Bottom Bourdon Tube/Tip Material: 316 stainless steel Bourdon Socket Material: 316 stainless steel Bellows/Socket Material: Blow-out Protection: Provide Case Material: Stainless steel / Cast iron / Phenolic Window Material: Glass Manufacturer: Ashcroft (or equal)			DIAPHRAGM SEAL	Diaphragm Material: 316 stainless steel	
					Bottom Housing Material: 316 stainless steel	
					Fill Fluid: Manufacturer's Recommendation	
					Process Connection: 1/2 inch NPT	
					Gauge Connection: 1/2 inch NPT	
					Manufacturer: Ashcroft (or equal)	
	EXCESS FLOW CHECK	Material: 316 stainless steel				
		Process Connection: 1/2 inch NPT				
		Gauge Connection: 1/2 inch NPT				
		Manufacturer: Chemiquip (or equal) ‡				
	PULSATION DAMPENER	Type: J. A. Campbell 'Microbean'				
		Material: 316 stainless steel				
		Process Connection: 1/2 inch NPT				
Gauge Connection: 1/2 inch NPT						
Manufacturer: Ashcroft (or equal)						
REMARKS	Provide accessories (Diaphragm Seal, Excess Flow Check, Pulsation Dampener) only when listed.					
	‡ Chemiquip Products Co., Inc. 524 64th Street West New York, New Jersey 07093 USA					
	DP : Design Pressure					
	DT : Design Temperature					
DISTRIBUTION	Tag Number	Service	Scale Range	Accessories	Installation	Comments
	PI-23-1504	Decomposer Acid Injection Pump P-2305A Discharge	0 to 6 kg/cm2(g)	Diaphragm Seal	6-101 Detail A	Maximum Temperature is 120°C
						DP : 4.921 kg/cm2(g) DT : 250 degC
	PI-23-1505	Decomposer Acid Injection Pump P-2305B Discharge	0 to 6 kg/cm2(g)	Diaphragm Seal	6-101 Detail A	Maximum Temperature is 120°C
						DP : 4.921 kg/cm2(g) DT : 250 degC
	PI-23-1601	Decomposer Acid Injection Pump P-2305C Discharge	0 to 6 kg/cm2(g)	Diaphragm Seal	6-101 Detail A	Maximum Temperature is 120°C
						DP : 4.921 kg/cm2(g) DT : 250 degC
	PI-23-1602	Decomposer Acid Injection Pump P-2305D Discharge	0 to 6 kg/cm2(g)	Diaphragm Seal	6-101 Detail A	Maximum Temperature is 120°C
						DP : 4.921 kg/cm2(g) DT : 250 degC
	PI-23-1603	Decomposer Acid Injection Pump P-2305E Discharge	0 to 6 kg/cm2(g)	Diaphragm Seal	6-101 Detail A	Maximum Temperature is 120°C
						DP : 4.921 kg/cm2(g) DT : 250 degC
PI-23-1604	Decomposer Acid Injection Pump P-2305F Discharge	0 to 6 kg/cm2(g)	Diaphragm Seal	6-101 Detail A	Maximum Temperature is 120°C	
					DP : 4.921 kg/cm2(g) DT : 250 degC	
§	§	to		6-101	§ Note 1	

ALL ITEMS SHALL COMPLY WITH THE GENERAL REQUIREMENTS SHOWN ON THE GENERAL REQUIREMENTS SECTION OF THIS SPECIFICATION.

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**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117							
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300							
REV	O1	A1	F1		Z1	DOCUMENT No. 14780-8120-PS-0032						
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014		MAY 10, 2016							
PREP. BY	S.W.Park	S.W.Park	S.W.Park		S.W.Park							
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung		J.S.Chung							
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee		S.W.Lee							
	1	General	Item No.	PI-23-0302		PI-23-0303		PI-23-0304		PI-23-0403		
P&ID No.			14780-8120-25-23-0003		14780-8120-25-23-0003		14780-8120-25-23-0003		14780-8120-25-23-0004			
			Preflash Column Feed Filter A Inlet		Preflash Column Feed Filter B Inlet		Preflash Column Feed Filters Outlet		Preflash Column Top			
Service												
	3	Line Size	inch		8"		8"		8"		V-2301	
	4	Fluid Properties	Type									
State			LIQ		LIQ		LIQ		VAP			
Description			HYDROCARBON		HYDROCARBON		HYDROCARBON		HYDROCARBON			
Corrosion/ Erosion due to												
	5											
	21	Operating temp °C	Minimum									
Normal			82		82		38		53			
Maximum												
	22	Operating Pressures kg/cm²g	Normal	0.96		0.96		0.07		-1.00		
	23		Maximum									
	24		Minimum									
	26	Instrument DP (Nor/Max) kg/cm²		Nil		Nil		Nil		Nil		
	32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	
	33	High Alarm kg/cm²g		N/A		N/A		N/A		N/A		
	34	Low Alarm kg/cm²g		N/A		N/A		N/A		N/A		
	35	Design Pressure kg/cm² (G)		3.5		3.5		3.5		3.5/FV		
	36	Design Temperature °C		130		130		120		195		
	37	Remarks										
	38	Revision										

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-0404		PI-23-0502		PI-23-0503		PI-23-0504
		P&ID No.	14780-8120-25-23-0004		14780-8120-25-23-0005		14780-8120-25-23-0005A		14780-8120-25-23-0005A
		Service	Preflash Column Bottom		S1.5 Steam to Preflash Column Steam Heater		Preflash Column Steam Heater Condensate Pump A		Preflash Column Steam Heater Condensate Pump B
3	Line Size	inch		V-2301	18"		3"		3"
4	Fluid Properties	Type							
		State	VAP		VAP		LIQ		LIQ
		Description	CUMENE		STEAM		CONDENSATE		CONDENSATE
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	64		135		135		135
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	-1.00		1.50		4.07		4.07
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	3.5/FV		7 / FV		9.233		9.233
36	Design Temperature	°C	195		195		195		195
37	Remarks								
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME:		PTT PHENOL COMPANY LIMITED				PROJECT No: 120117					
LOCATION:		MAP TA PHUT, THAILAND				UNIT No: 2300					
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032						
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016							
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park							
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung							
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee							
1	General	Item No.	PI-23-0601		PI-23-0702		PI-23-0703		PI-23-0704		
		P&ID No.	14780-8120-25-23-0006		14780-8120-25-23-0007		14780-8120-25-23-0007		14780-8120-25-23-0007A		
		Service	Concentration Cumene Quench Drum		S3 Steam to Flash Column Vaporizer		Sample Receiver Outlet		Flash Column Vaporizer Condensate Pump A		
3	Line Size	inch		D-2301		12"		3/4"		3"	
4	Fluid Properties	Type									
		State	VAP		VAP		LIQ		LIQ		
		Description	CUMENE		STEAM		HYDROCARBON		CONDENSATE		
		Corrosion/ Erosion due to									
5											
21	Operating temp °C	Minimum									
		Normal	38		135		70		135		
		Maximum									
22	Operating Pressures kg/cm ² g	Normal	0.06		2.30		-1.00		4.07		
Maximum											
Minimum											
26	Instrument DP (Nor/Max)	kg/cm ²		Nil		Nil		Nil		Nil	
32	Instrument Range (From/ To)	(NOTE 1)		(NOTE 1)		(NOTE 1)		(NOTE 1)		(NOTE 1)	
33	High Alarm	kg/cm ² g		N/A		N/A		N/A		N/A	
34	Low Alarm	kg/cm ² g		N/A		N/A		N/A		N/A	
35	Design Pressure	kg/cm ² (G)		3.5		10.5/FV		10/FV		9.233	
36	Design Temperature	°C		120		195		250		195	
37	Remarks			NOTE 2							
38	Revision										
NOTES											
1. Refer to local gauge data sheet (14780-8550-DS-0010).											
2. S-12 steam condition (max. 205 C) for steamout shall be considered.											
SHEET 27 OF 40											



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-0705		PI-23-0803		PI-23-0804		PI-23-0805
		P&ID No.	14780-8120-25-23-0007A		14780-8120-25-23-0008		14780-8120-25-23-0008		14780-8120-25-23-0008
		Service	Flash Column Vaporizer Condensate Pump B		Flash Column Top		Flash Column Bottom		Reliable Cooling Water from Flash Column Bottoms Cooler
3	Line Size	inch	3"		V-2302		V-2302		6"
4	Fluid Properties	Type							
		State	LIQ		VAP		VAP		LIQ
		Description	CONDENSATE		CUMENE		HYDROCARBON		RELIABLE COOLING WATER
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	135		51		95		43
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	4.07		-1.00		-1.00		2.50
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	9.233		3.5/FV		3.5/FV		10.5
36	Design Temperature	°C	195		195		195		120
37	Remarks								
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117						
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300						
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032						
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016							
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park							
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung							
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee							
1	General	Item No.	PI-23-0806		PI-23-0807		PI-23-0808		PI-23-0901		
		P&ID No.	14780-8120-25-23-0008A		14780-8120-25-23-0008B		14780-8120-25-23-0008B		14780-8120-25-23-0009		
		Service	Reliable Cooling Water from Preflash and Flash Column Condenser		Preflash and Flash Columns Overhead Pump A		Preflash and Flash Columns Overhead Pump B		Flash Column Bottoms Pump A		
3	Line Size	inch		28"		8"		8"		6"	
4	Fluid Properties	Type									
		State	LIQ		LIQ		LIQ		LIQ		
		Description	RELIABLE COOLING WATER		RELIABLE COOLING WATER		RELIABLE COOLING WATER		CUMENE HYDROPEROXIDE		
		Corrosion/ Erosion due to									
5											
21	Operating temp °C	Minimum									
		Normal	43		39		39		60		
		Maximum									
22	Operating Pressures kg/cm ² g	Normal	2.50		4.50		4.50		3.50		
23		Maximum									
24		Minimum									
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil		
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A		
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A		
35	Design Pressure	kg/cm ² (G)	10.5		13.55/FV		13.55/FV		10/FV		
36	Design Temperature	°C	120		120		120		195		
37	Remarks			NOTE 2		NOTE 2					
38	Revision										

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).
2. S-12 steam condition (max. 205 C) for steamout shall be considered.



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-0902		PI-23-1004		PI-23-1005		PI-23-1006
		P&ID No.	14780-8120-25-23-0009		14780-8120-25-23-0010		14780-8120-25-23-0010		14780-8120-25-23-0010
		Service	Flash Column Bottoms Pump B		Preflash and Flash Column Overhead Vapor to Flash Column Overhead Vapor		Chilled Water from Flash Column Overhead Vapor Chiller		Chilled Water from Concentration Vacuum Producing Equipment
3	Line Size	inch		6"	30"		6"		6"
4	Fluid Properties	Type							
		State	LIQ		VAP		LIQ		LIQ
		Description	CUMENE HYDROPEROXIDE		CUMENE		CHILLED WATER		CHILLED WATER
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	60		38		5		5
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	3.50		-1.00		2.50		2.50
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	10/FV		4.5/FV		11.5		11.5
36	Design Temperature	°C	195		120		120		120
37	Remarks			NOTE 2					
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).
2. S-12 steam condition (max. 205 C) for steamout shall be considered.



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-1007		PI-23-1102		PI-23-1203		PI-23-1204
		P&ID No.	14780-8120-25-23-0010		14780-8120-25-23-0011		14780-8120-25-23-0012A		14780-8120-25-23-0012A
		Service	Nitrogen Purge to Concentration Vacuum Producing Equipment		Cumene Flush Drum		Decomposer Feed Flush Drum		Process Water to Decomposer Feed
3	Line Size	inch		1"	2"		D-2303		1"
4	Fluid Properties	Type							
		State	VAP		VAP		VAP		LIQ
		Description	NITROGEN		HYDROCARBON		CUMENE		PROCESS WATER
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	AMB		38		38		50
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	0.84		0.70		0.70		5.40
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	10		10		10		10.5
36	Design Temperature	°C	120		120		120		120
37	Remarks			NOTE 2					
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).
2. S-12 steam condition (max. 205 C) for steamout shall be considered.



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME:		PTT PHENOL COMPANY LIMITED				PROJECT No: 120117						
LOCATION:		MAP TA PHUT, THAILAND				UNIT No: 2300						
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032							
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016								
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park								
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung								
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee								
1	General	Item No.	PI-23-1302		PI-23-1303		PI-23-1304		PI-23-1305			
		P&ID No.	14780-8120-25-23-0013		14780-8120-25-23-0013		14780-8120-25-23-0013		14780-8120-25-23-0013			
			Decomposer Drum		Decomposer Drum		Decomposer Product Pump A		Decomposer Product Pump B			
		Service										
3	Line Size	inch		3"		D-2304		6"		6"		
4	Fluid Properties	Type										
		State	VAP		VAP		LIQ		LIQ			
		Description	HYDROCARBON		HYDROCARBON		HYDROCARBON		HYDROCARBON			
		Corrosion/ Erosion due to										
5												
21	Operating temp °C	Minimum										
		Normal	60		60		75		75			
		Maximum										
22	Operating Pressures kg/cm ² g	Normal	0.70		0.70		5.90		5.90			
Maximum												
Minimum												
26	Instrument DP (Nor/Max)	kg/cm ²		Nil		Nil		Nil		Nil		
32	Instrument Range (From/ To)	(NOTE 1)		(NOTE 1)		(NOTE 1)		(NOTE 1)		(NOTE 1)		
33	High Alarm	kg/cm ² g		N/A		N/A		N/A		N/A		
34	Low Alarm	kg/cm ² g		N/A		N/A		N/A		N/A		
35	Design Pressure	kg/cm ² (G)		11/FV		11/FV		21		21		
36	Design Temperature	°C		250		250		250		250		
37	Remarks											
38	Revision											
NOTES												
1. Refer to local gauge data sheet (14780-8550-DS-0010).												
SHEET 32 OF 40												



CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-1402		PI-23-1403		PI-23-1404		PI-23-1405
		P&ID No.	14780-8120-25-23-0014A		14780-8120-25-23-0014A		14780-8120-25-23-0014		14780-8120-25-23-0014
		Service	Crude Product to Crude Product Cooler		Cooling Water from Crude Product Cooler		S4 Steam to Dehydrators		Dehydrator A Outlet
3	Line Size	inch	4"		10"		8"		4"
4	Fluid Properties	Type							
		State	LIQ		LIQ		VAP		LIQ
		Description	HYDROCARBON		COOLING WATER		STEAM		HYDROCARBON
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	140		43		160		140
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	4.90		2.50		4.00		4.90
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	21		7		16.5/FV		21
36	Design Temperature	°C	250		120		195		250
37	Remarks								
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-1406		PI-23-1407		PI-23-1702		PI-23-1703
		P&ID No.	14780-8120-25-23-0014		14780-8120-25-23-0014		14780-8120-25-23-0017		14780-8120-25-23-0017
		Service	Dehydrator B Outlet		Dehydrators Inlet		Decomposer Circulation Pump A		Decomposer Circulation Pump B
3	Line Size	inch	4"		6"		14"		14"
4	Fluid Properties	Type							
		State	LIQ		LIQ		LIQ		LIQ
		Description	HYDROCARBON		HYDROCARBON		HYDROCARBON		HYDROCARBON
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	140		75		72		72
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	4.90		5.90		4.64		4.64
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	21		21		16.5		16.5
36	Design Temperature	°C	250		250		250		250
37	Remarks								
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-1704		PI-23-1801		PI-23-1802		PI-23-1803
		P&ID No.	14780-8120-25-23-0017		14780-8120-25-23-0018		14780-8120-25-23-0018		14780-8120-25-23-0018
			Reliable Cooling Water from Decomposer Coolers		Circulating Decomposer Liquid to Calorimeter No. 1 Inlet		Circulating Decomposer Liquid from Calorimeter No. 1		Circulating Decomposer Liquid to Calorimeter No. 2 Inlet
		Service							
3	Line Size	inch		24"	3/4"		3/4"		3/4"
4	Fluid Properties	Type							
		State	LIQ		LIQ		LIQ		LIQ
		Description	RELIABLE COOLING WATER		HYDROCARBON		HYDROCARBON		HYDROCARBON
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	43		79		79		79
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	2.50		4.64		4.64		4.64
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	10.5		10.5		10.5		10.5
36	Design Temperature	°C	120		250		250		250
37	Remarks								
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-1804		PI-23-1903		PI-23-1904		PI-23-1905
		P&ID No.	14780-8120-25-23-0018		14780-8120-25-23-0019		14780-8120-25-23-0019		14780-8120-25-23-0019
		Service	Circulating Decomposer Liquid from Calorimeter No. 2		Direct Neutralization Static Mixers Inlet		Direct Neutralization Static Mixers Outlet		Neutralizing Agent Injection Pump A
3	Line Size	inch		3/4"	6"		6"		1/2"
4	Fluid Properties	Type							
		State	LIQ		LIQ		LIQ		LIQ
		Description	HYDROCARBON		HYDROCARBON		HYDROCARBON		DIAMINE
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	79		31		31		31
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	4.64		4.40		4.40		4.40
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A
35	Design Pressure	kg/cm ² (G)	10.5		21.5		21.5		6.546
36	Design Temperature	°C	250		250		250		120
37	Remarks								
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



**PRESSURE INSTRUMENT
PROCESS SPECIFICATION**

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED				PROJECT No: 120117						
LOCATION: MAP TA PHUT, THAILAND				UNIT No: 2300						
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032					
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016						
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park						
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung						
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee						
1	General	Item No.	PI-23-1906		PI-23-2001		PI-23-2002		PI-23-2003	
		P&ID No.	14780-8120-25-23-0019		14780-8120-25-23-0020		14780-8120-25-23-0020A		14780-8120-25-23-0020A	
		Service	Neutralizing Agent Injection Pump B		Acid Aromatics Knockout Drum		Acid Aromatics Sump Pump Outlet		Nitrogen Purge to Acid Aromatics Sump	
3	Line Size	inch	1/2"		D-2306		3"		1"	
4	Fluid Properties	Type								
		State	LIQ		VAP		LIQ		VAP	
		Description	DIAMINE		HYDROCARBON		HYDROCARBON		NITROGEN	
		Corrosion/ Erosion due to								
5										
21	Operating temp °C	Minimum								
		Normal	31		38		38		AMB	
		Maximum								
22	Operating Pressures kg/cm ² g	Normal	4.40		0.01		7.52		0.84	
Maximum										
24		Minimum								
26	Instrument DP (Nor/Max)	kg/cm ²	Nil		Nil		Nil		Nil	
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm	kg/cm ² g	N/A		N/A		N/A		N/A	
34	Low Alarm	kg/cm ² g	N/A		N/A		N/A		N/A	
35	Design Pressure	kg/cm ² (G)	6.546		3.5		13.06		10	
36	Design Temperature	°C	120		325		325		120	
37	Remarks									
38	Revision									
NOTES 1. Refer to local gauge data sheet (14780-8550-DS-0010).										
SHEET 37 OF 40										



PRESSURE INSTRUMENT PROCESS SPECIFICATION

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117				
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300				
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032				
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016					
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park					
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung					
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee					
1	General	Item No.	PI-23-0813		PI-23-0907		PI-23-0706		PI-23-2101
		P&ID No.	14780-8120-25-23-0001D		14780-8120-25-23-0001D		14780-8120-25-23-0001E		14780-8120-25-23-0021
		Service	SN-2301		SN-2303		SN-2310		AD Funnel
3	Line Size	inch	3/4		3/4		3/4		
4	Fluid Properties	Type							
		State	LIQ		LIQ		LIQ		LIQ
		Description	CUMENE		CHP		HYDROCARBON		HYDROCARBON
		Corrosion/ Erosion due to							
5									
21	Operating temp °C	Minimum							
		Normal	39		60		70		AMB
		Maximum							
22	Operating Pressures kg/cm ² g	Normal	4.50		3.50		-1.00		0.20
23		Maximum							
24		Minimum							
26	Instrument DP (Nor/Max) kg/cm ²		Nil		Nil		Nil		Nil
32	Instrument Range (From/ To)		(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)	(NOTE 1)
33	High Alarm		N/A		N/A		N/A		N/A
34	Low Alarm		N/A		N/A		N/A		N/A
35	Design Pressure kg/cm ² (G)		13.55 / FV		8.86 / FV		4 / FV		3.5
36	Design Temperature °C		120		250		250		120
37	Remarks								
38	Revision								

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).



PRESSURE INSTRUMENT PROCESS SPECIFICATION

CUSTOMERS NAME: PTT PHENOL COMPANY LIMITED					PROJECT No: 120117					
LOCATION: MAP TA PHUT, THAILAND					UNIT No: 2300					
REV	O1	A1	F1	Z1	DOCUMENT No. 14780-8120-PS-0032					
DATE	DEC 30, 2013	APR 18, 2014	JUN 25, 2014	MAY 10, 2016						
PREP. BY	S.W.Park	S.W.Park	S.W.Park	S.W.Park						
REVIEW BY	J.S.Chung	J.S.Chung	J.S.Chung	J.S.Chung						
APP. BY	S.W.Lee	S.W.Lee	S.W.Lee	S.W.Lee						
1	General	Item No.	PI-23-2301							
		P&ID No.	14780-8120-25-23-0023							
		Service	CHD Funnel							
3	Line Size	inch								
4	Fluid Properties	Type								
		State	LIQ							
		Description	HYDROCARBON							
		Corrosion/ Erosion due to								
5										
21	Operating temp °C	Minimum								
		Normal	AMB							
		Maximum								
22	Operating Pressures kg/cm ² g	Normal	0.20							
Maximum										
24		Minimum								
26	Instrument DP (Nor/Max)	kg/cm ²		Nil						
32	Instrument Range (From/ To)			(NOTE 1)	(NOTE 1)					
33	High Alarm			N/A						
34	Low Alarm			N/A						
35	Design Pressure	kg/cm ² (G)		3.5						
36	Design Temperature	°C		120						
37	Remarks									
38	Revision									

NOTES

1. Refer to local gauge data sheet (14780-8550-DS-0010).

Revision
Indication

Note: The information in this document is confidential and the property of UOP LLC and must not be disclosed to others or reproduced in any manner or used for any purpose whatsoever without its written permission.

A Honeywell Company



PRESSURE GAUGES

PROJECT SPECIFICATION

963766 — 602-N

REV	DATE	BY	APVD	REV	DATE	BY	APVD
0	25JUN12	MVD	MW				

Notes:

- Pressure gauges shown without tag numbers on the UOP P&I Diagrams shall be specified by the Contractor in accordance with the general procedure guidelines shown below.

Pressure Gauge Range Selection Guidelines:

English Units		Metric and SI Units			
psig		kg/cm ² (g) or bar (g)		kPa (g) §	
Operating Pressure	Gauge Range	Operating Pressure	Gauge Range	Operating Pressure	Gauge Range
Vacuum < P ≤ 3	-30"Hg/ 0 / +15	Vacuum < P ≤ 0.2	-1/ 0 / +1.5	Vacuum < P ≤ 20	-100/ 0 / +150
3 < P ≤ 10	0 - 15	0.2 < P ≤ 0.6	0 - 1.0	20 < P ≤ 60	0 - 100
10 < P ≤ 20	0 - 30	0.6 < P ≤ 1.0	0 - 1.6	60 < P ≤ 100	0 - 160
20 < P ≤ 40	0 - 60	1.0 < P ≤ 1.5	0 - 2.5	100 < P ≤ 150	0 - 250
40 < P ≤ 65	0 - 100	1.5 < P ≤ 2.5	0 - 4	150 < P ≤ 250	0 - 400
65 < P ≤ 90	0 - 160	2.5 < P ≤ 4.0	0 - 6	250 < P ≤ 400	0 - 600
90 < P ≤ 120	0 - 200	4.0 < P ≤ 6.5	0 - 10	400 < P ≤ 650	0 - 1000
120 < P ≤ 180	0 - 300	6.5 < P ≤ 10	0 - 16	650 < P ≤ 1000	0 - 1600
180 < P ≤ 250	0 - 400	10 < P ≤ 15	0 - 25	1000 < P ≤ 1500	0 - 2500
250 < P ≤ 350	0 - 600	15 < P ≤ 25	0 - 40	1500 < P ≤ 2500	0 - 4000
350 < P ≤ 450	0 - 800	25 < P ≤ 40	0 - 60	2500 < P ≤ 4000	0 - 6000
450 < P ≤ 600	0 - 1000	40 < P ≤ 65	0 - 100	4000 < P ≤ 6500	0 - 10000
600 < P ≤ 900	0 - 1500	65 < P ≤ 100	0 - 160	6500 < P ≤ 10000	0 - 16000
900 < P ≤ 1200	0 - 2000	100 < P ≤ 150	0 - 250	10000 < P ≤ 15000	0 - 25000
1200 < P ≤ 1900	0 - 3000	150 < P ≤ 240	0 - 400	15000 < P ≤ 24000	0 - 40000
1900 < P ≤ 3500	0 - 5000	240 < P ≤ 420	0 - 600	24000 < P ≤ 42000	0 - 60000

§ Pressure gauge selection guidelines for MPa (g) units are obtainable by dividing the kPa (g) values listed above by 1000.

- Provide compound gauge with appropriate upper range value on discharge of any pump taking suction from a vessel operating under a vacuum.
- Provide compound gauge at any ejector inlet.
- Provide a pulsation dampener on any gauge associated with a reciprocating compressor or with a reciprocating or proportioning pump.
- Provide an excess flow check valve on inlet of any gauge operating above 1000 psig or 70 kg/cm² (g) or 70 bar (g) or 7000 kPa (g) or 7 MPa (g).